PlantFolio — Database Design Document

Oct 31, 2026 · @ranshitha

Overview

PlantFolio needs **14 MySQL tables**: the 10 from the proposal plus 4 that the built pages already rely on (`plant_photos` , `plant_likes` , `listing_photos` , `disease_guides`). All use InnoDB and utf8mb4, and PHP reaches them through PDO prepared statements.

This design was built from the live preview (feature/pages-ui build), `includes/mock-data.php` (whose arrays already use the future column names), `database/README.md` , `ml-service/README.md` and the form fields of every page in `public/` .

Module	Tables	Pages that use them
Accounts	`users`	P02 Sign up / Log in, P14 Public Profile, P15 Settings
My garden	`collections` , `plants` , `plant_species` , `plant_photos` , `health_logs` , `care_reminders`	P03 Dashboard, P04 My Garden, P05 Plant Profile, P06 Add / Edit Plant
Community	`plant_likes`	P01 Landing, P09 Explore, P14 Public Profile
Exchange	`exchange_listings` , `listing_photos` , `listing_interests` , `user_ratings`	P10 Marketplace, P11 Listing Detail, P12 Create Listing, P13 My Exchanges
AI tools	`ml_predictions` , `disease_guides`	P07 Plant Identifier, P08 Health Check

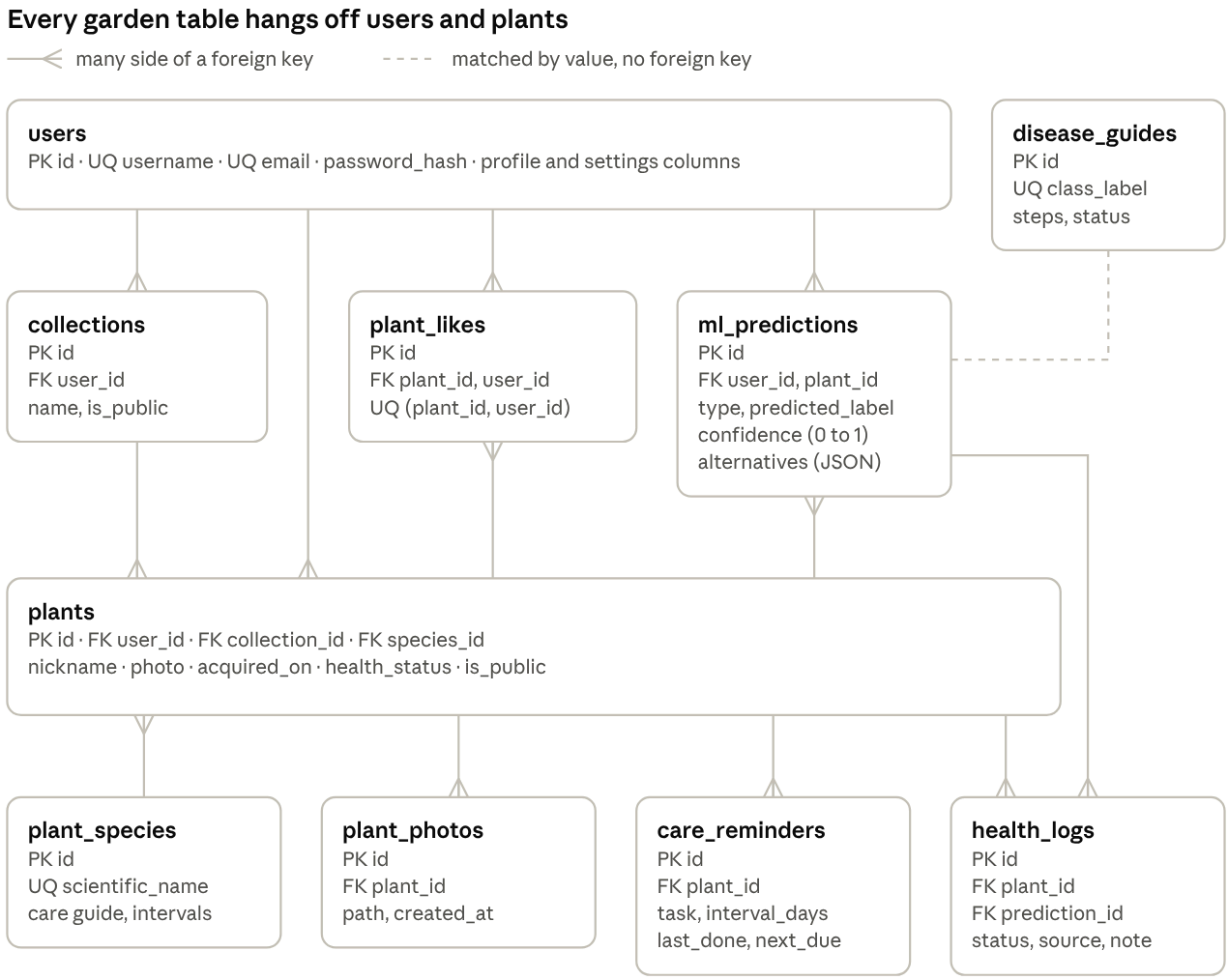
Conventions used throughout

- Plural snake_case table names, an `id INT UNSIGNED AUTO_INCREMENT` primary key, and `<table>_id` foreign keys.
- Photos are files in `public/uploads/` ; the database stores only the path.
- Health status everywhere is one of `thriving` , `stable` , `needs_attention` , `sick` .

- Counts, averages and distances (`plant_count` , `rating_avg` , `interest_count` , `like_count`) are never stored. Queries calculate them with `COUNT()` and `AVG()` , so they can't go out of date.
- Columns marked **(proposed)** below are not in `database/README.md` yet. They come from a page that needs them and should be confirmed in task DES-07.

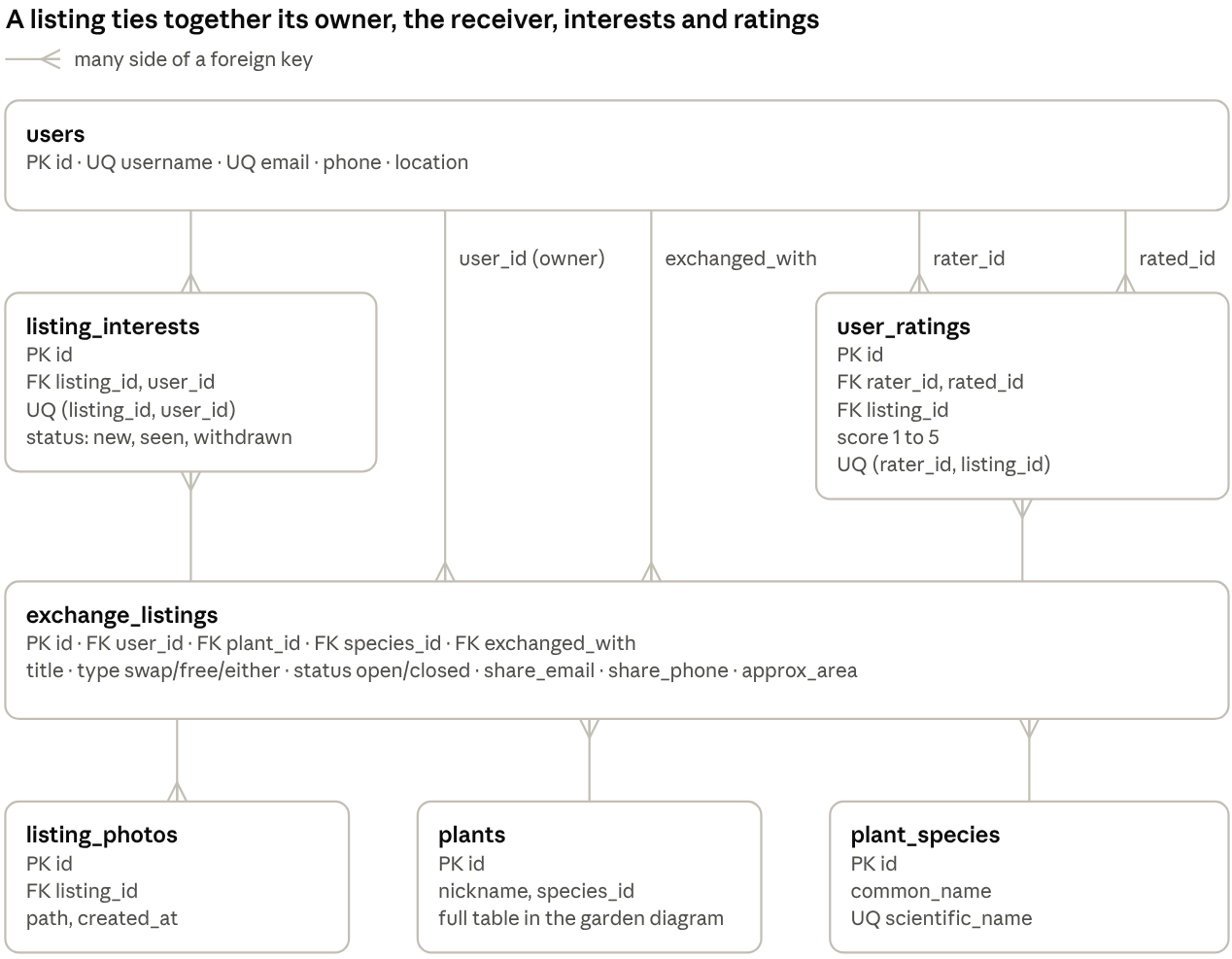
Entity-relationship diagram

`users` and `plants` are the two hubs: 10 of the 14 tables point at one or both. The diagram is split in two so each part stays readable; `users` , `plants` and `plant_species` appear in both.



ER diagram, part 1 · garden and AI tools · 10 tables

Every garden table belongs to a plant, and a plant belongs to a member. `disease_guides` is linked to results by matching the class name, not by a foreign key.



ER diagram, part 2 · exchange · 7 tables

A listing has two links to `users` (its owner and, after the swap, the receiver), and a rating has two more (giver and receiver). A listing uses `species_id` only when it offers cuttings or seedlings that have no row in `plants`.

Schema design

The full schema is 14 tables with 119 columns, 14 primary keys, 22 foreign keys, 9 unique keys and 3 CHECK constraints. This section is the logical design; the column-by-column detail follows in Table structures, and the runnable version is the SQL schema script.

Schema at a glance

#	Table	Module	Columns	Primary key	Foreign keys	Unique keys
1	users	Accounts	16	id	—	username; email
2	collections	My garden	5	id	user_id	(user_id, name)
3	plant_species	My garden	11	id	—	scientific_name
4	plants	My garden	10	id	user_id, collection_id, species_id	—
5	plant_photos	My garden	4	id	plant_id	—
6	health_logs	My garden	8	id	plant_id, prediction_id	—
7	care_reminders	My garden	6	id	plant_id	(plant_id, task)
8	plant_likes	Community	4	id	plant_id, user_id	(plant_id, user_id)
9	ml_predictions	AI tools	12	id	user_id, plant_id	—
10	disease_guides	AI tools	7	id	—	class_label
11	exchange_listings	Exchange	19	id	user_id, plant_id, species_id, exchanged_with	—
12	listing_photos	Exchange	4	id	listing_id	—
13	listing_interests	Exchange	6	id	listing_id, user_id	(listing_id, user_id)
14	user_ratings	Exchange	7	id	rater_id, rated_id, listing_id	(rater_id, listing_id)

Relational schema

Primary keys are in **bold**, foreign keys in *italics*.

- **users** (**id**, username, full_name, email, phone, password_hash, bio, location, avatar, new_plants_public, show_listing_area, dashboard_reminders, email_reminders, failed_logins, locked_until, created_at)
- **collections** (**id**, *user_id* → users, name, is_public, created_at)
- **plant_species** (**id**, common_name, scientific_name, photo, light, water, humidity, pet_safe, water_interval_days, fertilise_interval_days, notes)
- **plants** (**id**, *user_id* → users, *collection_id* → collections, *species_id* → plant_species, nickname, photo, acquired_on, health_status, is_public, created_at)

- **plant_photos** (**id**, *plant_id* → plants, path, created_at)
- **health_logs** (**id**, *plant_id* → plants, *prediction_id* → ml_predictions, status, source, note, photo, logged_at)
- **care_reminders** (**id**, *plant_id* → plants, task, interval_days, last_done, next_due)
- **plant_likes** (**id**, *plant_id* → plants, *user_id* → users, created_at)
- **ml_predictions** (**id**, *user_id* → users, *plant_id* → plants, type, predicted_label, confidence, alternatives, image_path, upload_name, upload_size, model_version, created_at)
- **disease_guides** (**id**, class_label, crop, cause, symptoms, steps, status)
- **exchange_listings** (**id**, *user_id* → users, *plant_id* → plants, *species_id* → plant_species, title, description, type, looking_for, location, size, plant_condition, photo, share_email, share_phone, approx_area, status, *exchanged_with* → users, exchanged_at, created_at)
- **listing_photos** (**id**, *listing_id* → exchange_listings, path, created_at)
- **listing_interests** (**id**, *listing_id* → exchange_listings, *user_id* → users, message, status, created_at)
- **user_ratings** (**id**, *rater_id* → users, *rated_id* → users, *listing_id* → exchange_listings, score, comment, created_at)

Domains (allowed values)

Domain	MySQL type	Allowed values	Used by
Health status	ENUM	thriving, stable, needs_attention, sick	plants.health_status, health_logs.status, disease_guides.status
Log source	ENUM	manual, ml	health_logs.source
Reminder task	ENUM	water, fertilise, recheck	care_reminders.task
Prediction type	ENUM	species, disease	ml_predictions.type
Listing type	ENUM	swap, free, either	exchange_listings.type
Listing status	ENUM	open, closed	exchange_listings.status

Domain	MySQL type	Allowed values	Used by
Interest status	ENUM	new, seen, withdrawn	listing_interests.status
Yes/no flag	TINYINT(1)	0 or 1	is_public, pet_safe, share_email, share_phone, approx_area, the four Settings flags
Confidence	DECIMAL(5,4)	0.0000 to 1.0000	ml_predictions.confidence
Rating score	TINYINT UNSIGNED	1 to 5	user_ratings.score
Interval	SMALLINT UNSIGNED	1 or more days	care_reminders.interval_days, plant_species interval columns
File path	VARCHAR(255)	Relative path under public/uploads/	every photo, avatar, path and image_path column

Constraints enforced by the database

Constraint	Type	Table	Rule
uq_users_username	UNIQUE	users	No two members share a username
uq_users_email	UNIQUE	users	No two members share an email
uq_collections_user_name	UNIQUE	collections	A member's collection names are distinct
uq_species_scientific	UNIQUE	plant_species	One row per scientific name
uq_plant_likes	UNIQUE	plant_likes	One like per member per plant
uq_reminders_plant_task	UNIQUE	care_reminders	One reminder per task per plant
uq_disease_guides_label	UNIQUE	disease_guides	One guide per model class
uq_interests_listing_user	UNIQUE	listing_interests	One interest row per member per listing

Constraint	Type	Table	Rule
uq_ratings_rater_listing	UNIQUE	user_ratings	One rating per member per exchange
chk_reminders_interval	CHECK	care_reminders	interval_days > 0
chk_predictions_confidence	CHECK	ml_predictions	confidence BETWEEN 0 AND 1
chk_ratings_score	CHECK	user_ratings	score BETWEEN 1 AND 5

All 22 foreign keys and their ON DELETE rules are listed under Relationships.

Business rules enforced in PHP

These rules span rows or tables, so SQL constraints cannot check them; the PHP that writes the row must.

Rule	Where it is checked
A member can only edit or delete rows they own (`user_id` = session user)	Every UPDATE and DELETE, in the WHERE clause
`next_due` always equals `last_done + interval_days`	Add Plant form and Mark done
`plants.health_status` equals the status of the plant's newest health log	Same transaction as the health_logs INSERT
A listing has `plant_id` or `species_id`, not both	Create Listing form
`exchanged_with` and `exchanged_at` are set only when `status` = closed	Mark as exchanged
A rating needs a closed listing, and the rater must be its owner or `exchanged_with`	Submit rating on My Exchanges
`rated_id` differs from `rater_id`	Submit rating
Private plants and collections (`is_public` = 0) are never returned to other members	Every public query adds `is_public = 1`

Rule	Where it is checked
Contact details are shown only after "I'm interested", and only the fields with share_email or share_phone = 1	Listing Detail query
An account locks for 15 minutes after 5 failed logins	Login (failed_logins, locked_until)

Optional tables (not in schema.sql yet)

Two open questions at the end of this doc would each add one table. They are designed here so DES-07 can accept or drop them.

Table	Columns	Needed for
towns	**id**, name (UQ), latitude DECIMAL(9,6), longitude DECIMAL(9,6)	The "3 km" distance on listing cards
remember_tokens	**id**, *user_id* → users (CASCADE), selector CHAR(24) UQ, token_hash CHAR(64), expires_at DATETIME	A secure "Remember me" login

Table structures

Every table below lists each column with its MySQL type, whether it can be NULL, its default and its key. PK = primary key, FK = foreign key, UQ = unique.

1. users

One row per member: login details, public profile and Settings choices.

Column	Type	Null	Default	Key	Notes
id	INT UNSIGNED	No	AUTO_INCREMENT	PK	
username	VARCHAR(30)	No		UQ	Handle shown as @nimali.grows
full_name	VARCHAR(100)	No			
email	VARCHAR(255)	No		UQ	Never shown publicly

Column	Type	Null	Default	Key	Notes
phone	VARCHAR(20)	Yes	NULL		Optional; shared only through a listing's share_phone
password_hash	VARCHAR(255)	No			From `password_hash()`
bio	VARCHAR(500)	Yes	NULL		
location	VARCHAR(100)	Yes	NULL		Home town, e.g. Kurunegala
avatar	VARCHAR(255)	Yes	NULL		Path in uploads; NULL shows initials
new_plants_public	TINYINT(1)	No	1		Settings: new plants start public (proposed)
show_listing_area	TINYINT(1)	No	1		Settings: default for a listing's approx_area (proposed)
dashboard_reminders	TINYINT(1)	No	1		Settings: show care tasks on the Dashboard (proposed)
email_reminders	TINYINT(1)	No	0		Settings toggle, disabled in the UI for now (proposed)
failed_logins	TINYINT UNSIGNED	No	0		Reset to 0 on a good login
locked_until	DATETIME	Yes	NULL		Set for 15 min after the 5th failure
created_at	DATETIME	No	CURRENT_TIMESTAMP		"Member since Mar 2025"

2. collections

A member's named groups of plants, such as Living room or Balcony.

Column	Type	Null	Default	Key	Notes
id	INT UNSIGNED	No	AUTO_INCREMENT	PK	

Column	Type	Null	Default	Key	Notes
user_id	INT UNSIGNED	No		FK → users.id	ON DELETE CASCADE
name	VARCHAR(60)	No		UQ (user_id, name)	No two collections with the same name per member
is_public	TINYINT(1)	No	1		0 shows a lock and hides it from others
created_at	DATETIME	No	CURRENT_TIMESTAMP		

3. plant_species

Reference data: the care guide for each species. Loaded from `database/species.csv` (at least 30 rows, including every class the recognition model predicts).

Column	Type	Null	Default	Key	Notes
id	INT UNSIGNED	No	AUTO_INCREMENT	PK	
common_name	VARCHAR(100)	No			Swiss cheese plant
scientific_name	VARCHAR(150)	No		UQ	Monstera deliciosa; matches the model's species labels
photo	VARCHAR(255)	Yes	NULL		Reference picture beside the Identifier's top match
light	VARCHAR(100)	Yes	NULL		Bright, indirect
water	VARCHAR(150)	Yes	NULL		When the top 5 cm of soil are dry
humidity	VARCHAR(60)	Yes	NULL		60%+ preferred
pet_safe	TINYINT(1)	Yes	NULL		NULL = unknown
water_interval_days	SMALLINT UNSIGNED	No	7		Pre-fills the Add Plant reminder
fertilise_interval_days	SMALLINT UNSIGNED	No	30		Pre-fills the Add Plant reminder
notes	TEXT	Yes	NULL		Extra care tips

4. plants

Each individual plant profile.

Column	Type	Null	Default	Key	Notes
id	INT UNSIGNED	No	AUTO_INCREMENT	PK	
user_id	INT UNSIGNED	No		FK → users.id	ON DELETE CASCADE
collection_id	INT UNSIGNED	Yes	NULL	FK → collections.id	ON DELETE SET NULL: deleting a collection keeps its plants
species_id	INT UNSIGNED	Yes	NULL	FK → plant_species.id	ON DELETE SET NULL; NULL when the species is unknown
nickname	VARCHAR(60)	No			Monty
photo	VARCHAR(255)	Yes	NULL		Main card photo
acquired_on	DATE	Yes	NULL		
health_status	ENUM('thriving','stable','needs_attention','sick')	No	'stable'	Index	Copied from the newest health log
is_public	TINYINT(1)	No	1		0 = only the owner can see it
created_at	DATETIME	No	CURRENT_TIMESTAMP		

5. plant_photos

Extra photos for the Plant Profile gallery (optional table).

Column	Type	Null	Default	Key	Notes
id	INT UNSIGNED	No	AUTO_INCREMENT	PK	
plant_id	INT UNSIGNED	No		FK → plants.id	ON DELETE CASCADE
path	VARCHAR(255)	No			
created_at	DATETIME	No	CURRENT_TIMESTAMP		Gallery shows newest first

6. health_logs

The dated plant story on the Plant Profile. Each row is a manual update or a saved AI result.

Column	Type	Null	Default	Key	Notes
id	INT UNSIGNED	No	AUTO_INCREMENT	PK	
plant_id	INT UNSIGNED	No		FK → plants.id	ON DELETE CASCADE
prediction_id	INT UNSIGNED	Yes	NULL	FK → ml_predictions.id	Set when source = 'ml'; ON DELETE SET NULL (proposed)

Column	Type	Null	Default	Key	Notes
status	ENUM('thriving','stable','needs_attention','sick')	No			
source	ENUM('manual','ml')	No	'manual'		
note	VARCHAR(500)	Yes	NULL		Brown rings on the lower leaves.
photo	VARCHAR(255)	Yes	NULL		
logged_at	DATETIME	No	CURRENT_TIMESTAMP	Index (plant_id, logged_at)	Story is shown newest first

7. care_reminders

Recurring care tasks. A switched-off reminder on the Add Plant form simply has no row.

Column	Type	Null	Default	Key	Notes
id	INT UNSIGNED	No	AUTO_INCREMENT	PK	
plant_id	INT UNSIGNED	No		FK → plants.id	ON DELETE CASCADE
task	ENUM('water','fertilise','recheck')	No		UQ (plant_id, task)	`recheck` is the one-off "re-check in 7 days" after a Health Check (proposed)
interval_days	SMALLINT UNSIGNED	No			CHECK > 0
last_done	DATE	Yes	NULL		NULL until first done
next_due	DATE	No		Index	Always last_done + interval_days

8. plant_likes

Likes on public plants (optional table, COM-15). One like per member per plant.

Column	Type	Null	Default	Key	Notes
id	INT UNSIGNED	No	AUTO_INCREMENT	PK	
plant_id	INT UNSIGNED	No		FK → plants.id, UQ (plant_id, user_id)	ON DELETE CASCADE

Column	Type	Null	Default	Key	Notes
user_id	INT UNSIGNED	No		FK → users.id	ON DELETE CASCADE
created_at	DATETIME	No	CURRENT_TIMESTAMP		

9. ml_predictions

Every species recognition or disease diagnosis, saved or not, so model accuracy can be measured for the report.

Column	Type	Null	Default	Key	Notes
id	INT UNSIGNED	No	AUTO_INCREMENT	PK	Used in `identify.php?prediction=5`
user_id	INT UNSIGNED	No		FK → users.id	ON DELETE CASCADE
plant_id	INT UNSIGNED	Yes	NULL	FK → plants.id	NULL until attached to a plant; ON DELETE SET NULL
type	ENUM('species','disease')	No			
predicted_label	VARCHAR(150)	No		Index	Top result: a scientific name, or a class such as `Tomato___Early_blight`
confidence	DECIMAL(5,4)	No			0.0000–1.0000, CHECK in range; High ≥ 0.80, Not sure < 0.60
alternatives	JSON	Yes	NULL		The two runners-up from Flask's top 3
image_path	VARCHAR(255)	No			
upload_name	VARCHAR(255)	Yes	NULL		Original file name shown on the result page (proposed)
upload_size	INT UNSIGNED	Yes	NULL		Bytes (proposed)
model_version	VARCHAR(50)	No			species-mobilenetv2-v1
created_at	DATETIME	No	CURRENT_TIMESTAMP		

10. disease_guides

What the Health Check says about each disease class (proposed table; could instead be a JSON file beside the model).

Column	Type	Null	Default	Key	Notes
id	INT UNSIGNED	No	AUTO_INCREMENT	PK	

Column	Type	Null	Default	Key	Notes
class_label	VARCHAR(100)	No		UQ	Same text as ml_predictions.predicted_label
crop	VARCHAR(50)	No			Tomato
cause	VARCHAR(255)	Yes	NULL		Caused by the fungus Alternaria solani
symptoms	TEXT	Yes	NULL		Plain-language description
steps	JSON	Yes	NULL		Array of the four care steps
status	ENUM('thriving','stable','needs_attention','sick')	No			The health_status a saved result gives the plant

11. exchange_listings

Plants offered for swap or free. A listing either points at one of the owner's plants (`plant_id`) or names its own species for cuttings and seedlings (`species_id`).

Column	Type	Null	Default	Key	Notes
id	INT UNSIGNED	No	AUTO_INCREMENT	PK	
user_id	INT UNSIGNED	No		FK → users.id	Owner; ON DELETE CASCADE
plant_id	INT UNSIGNED	Yes	NULL	FK → plants.id	ON DELETE SET NULL
species_id	INT UNSIGNED	Yes	NULL	FK → plant_species.id	Only when plant_id is NULL; ON DELETE SET NULL
title	VARCHAR(100)	No			Pothos 'Golden', 3 rooted cuttings
description	TEXT	Yes	NULL		
type	ENUM('swap','free','either')	No	'swap'		
looking_for	VARCHAR(150)	Yes	NULL		succulents or a snake plant pup
location	VARCHAR(100)	No			Pick-up town; can differ from users.location
size	VARCHAR(60)	Yes	NULL		3 cuttings, 15 to 20 cm
plant_condition	VARCHAR(100)	Yes	NULL		Not `condition`, a MySQL reserved word
photo	VARCHAR(255)	No			Card photo
share_email	TINYINT(1)	No	1		Owner's choice after "I'm interested"
share_phone	TINYINT(1)	No	0		

Column	Type	Null	Default	Key	Notes
approx_area	TINYINT(1)	No	1		Show town only, never an address
status	ENUM('open','closed')	No	'open'	Index (status, created_at)	
exchanged_with	INT UNSIGNED	Yes	NULL	FK → users.id	Set by Mark as exchanged; ON DELETE SET NULL
exchanged_at	DATETIME	Yes	NULL		NULL while open
created_at	DATETIME	No	CURRENT_TIMESTAMP		"2 days ago", Newest sort

12. listing_photos

Extra photos for the Listing Detail gallery, shown after the listing's own photo.

Column	Type	Null	Default	Key	Notes
id	INT UNSIGNED	No	AUTO_INCREMENT	PK	
listing_id	INT UNSIGNED	No		FK → exchange_listings.id	ON DELETE CASCADE
path	VARCHAR(255)	No			
created_at	DATETIME	No	CURRENT_TIMESTAMP		Oldest first

13. listing_interests

"I'm interested" requests. One row per member per listing; withdrawing changes the status instead of deleting.

Column	Type	Null	Default	Key	Notes
id	INT UNSIGNED	No	AUTO_INCREMENT	PK	
listing_id	INT UNSIGNED	No		FK → exchange_listings.id, UQ (listing_id, user_id)	ON DELETE CASCADE
user_id	INT UNSIGNED	No		FK → users.id	The requester; ON DELETE CASCADE
message	VARCHAR(500)	Yes	NULL		
status	ENUM('new','seen','withdrawn')	No	'new'		Withdrawn rows are not counted

Column	Type	Null	Default	Key	Notes
created_at	DATETIME	No	CURRENT_TIMESTAMP		

14. user_ratings

Trust ratings after a finished exchange. Both members may rate each other once.

Column	Type	Null	Default	Key	Notes
id	INT UNSIGNED	No	AUTO_INCREMENT	PK	
rater_id	INT UNSIGNED	No		FK → users.id, UQ (rater_id, listing_id)	Who gave it; ON DELETE CASCADE
rated_id	INT UNSIGNED	No		FK → users.id	Who received it; PHP checks it differs from rater_id
listing_id	INT UNSIGNED	No		FK → exchange_listings.id	The finished exchange; ON DELETE CASCADE
score	TINYINT UNSIGNED	No			CHECK 1–5
comment	VARCHAR(500)	Yes	NULL		
created_at	DATETIME	No	CURRENT_TIMESTAMP		Newest review first

Relationships

The schema has 22 foreign keys. Deleting a member removes everything they own (CASCADE); deleting an optional parent, such as a collection or species, only clears the link (SET NULL).

Foreign key	References	Cardinality	ON DELETE
collections.user_id	users.id	One user has many collections	CASCADE

Foreign key	References	Cardinality	ON DELETE
plants.user_id	users.id	One user has many plants	CASCADE
plants.collection_id	collections.id	One collection has 0..many plants	SET NULL
plants.species_id	plant_species.id	One species describes 0..many plants	SET NULL
plant_photos.plant_id	plants.id	One plant has 0..many photos	CASCADE
plant_likes.plant_id	plants.id	One plant has 0..many likes	CASCADE
plant_likes.user_id	users.id	One user gives 0..many likes	CASCADE
health_logs.plant_id	plants.id	One plant has 1..many log entries	CASCADE
health_logs.prediction_id	ml_predictions.id	One prediction is saved as 0..1 log entry	SET NULL
care_reminders.plant_id	plants.id	One plant has 0..3 reminders (one per task)	CASCADE
ml_predictions.user_id	users.id	One user runs many predictions	CASCADE
ml_predictions.plant_id	plants.id	One plant has 0..many predictions	SET NULL
exchange_listings.user_id	users.id	One user owns many listings	CASCADE

Foreign key	References	Cardinality	ON DELETE
exchange_listings.plant_id	plants.id	One plant is offered in 0..many listings	SET NULL
exchange_listings.species_id	plant_species.id	Used only when plant_id is NULL	SET NULL
exchange_listings.exchanged_with	users.id	The member who received the plant	SET NULL
listing_photos.listing_id	exchange_listings.id	One listing has 0..many extra photos	CASCADE
listing_interests.listing_id	exchange_listings.id	One listing gets 0..many interests	CASCADE
listing_interests.user_id	users.id	One user sends 0..many interests	CASCADE
user_ratings.rater_id / rated_id	users.id	Two FKs to users: giver and receiver	CASCADE
user_ratings.listing_id	exchange_listings.id	One finished listing gets 0..2 ratings	CASCADE

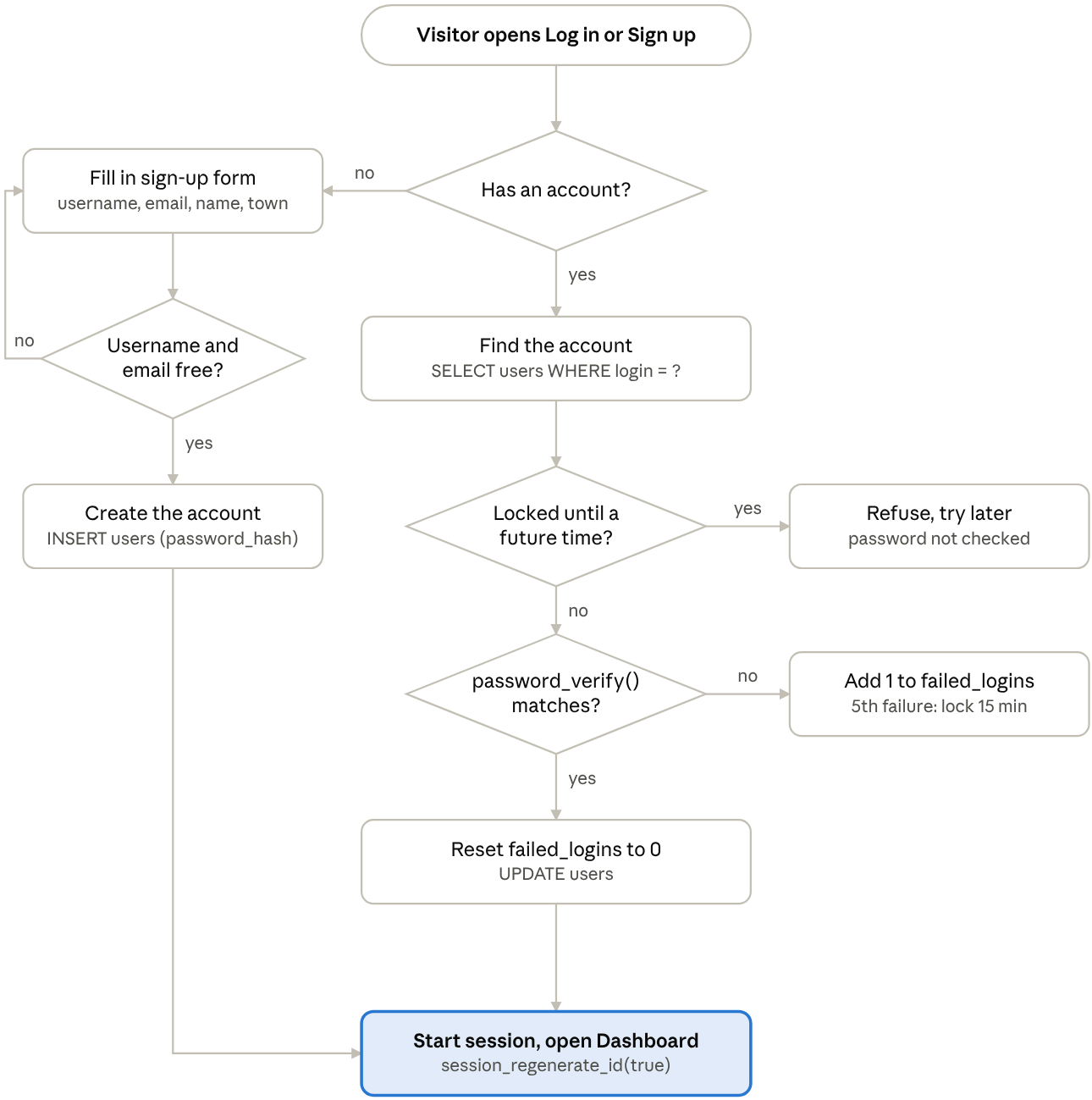
`disease_guides` has no foreign key: a Health Check result finds its guide by matching `ml_predictions.predicted_label` to `disease_guides.class_label`.

Flowcharts

The five main journeys below show which table each step reads or writes. The highlighted box is where each journey ends.

1. Sign up and log in

Five wrong passwords lock an account for 15 minutes

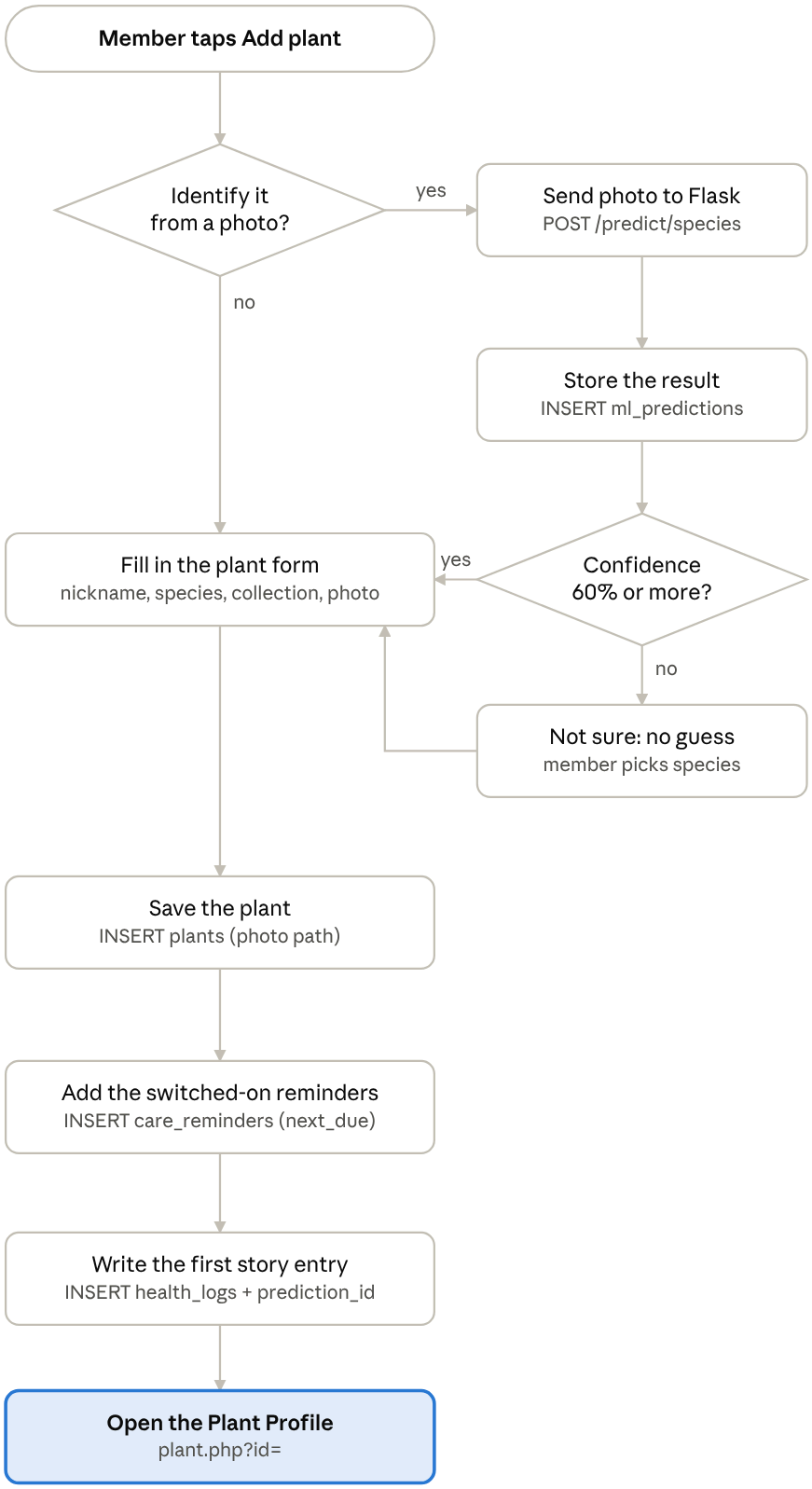


Flowchart 1 · sign up and log in

The lock check comes before the password check, so a locked account never reveals whether the password was right. Every failure shows the same generic error.

2. Add a plant

A new plant writes up to four tables in one save

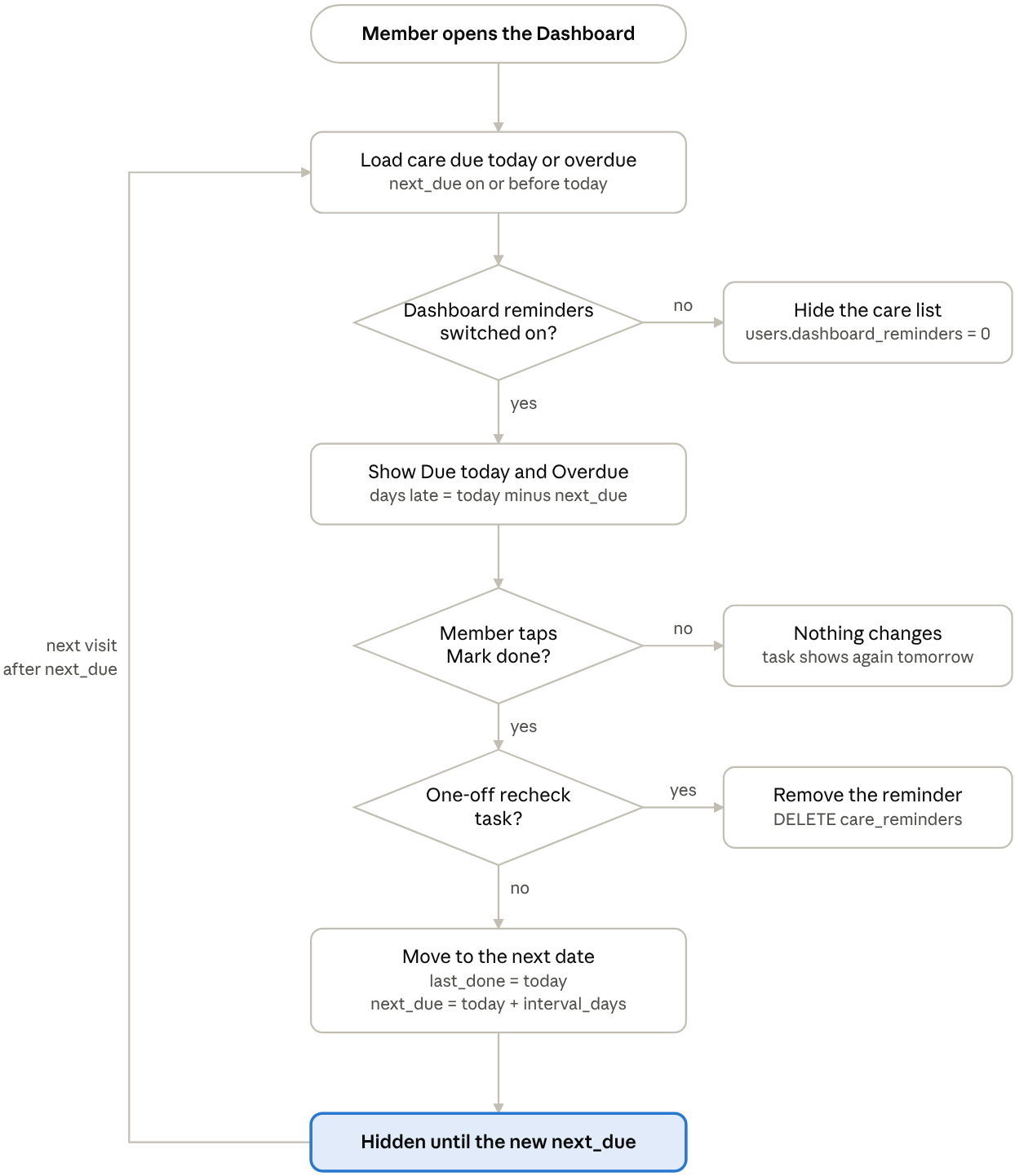


Flowchart 2 · add a plant with AI identification

The three inserts after the form (plants, care_reminders, health_logs) run in one transaction, then `ml_predictions.plant_id` is set so the result is linked to the new plant.

3. Care reminders

A reminder moves forward one interval each time it is done

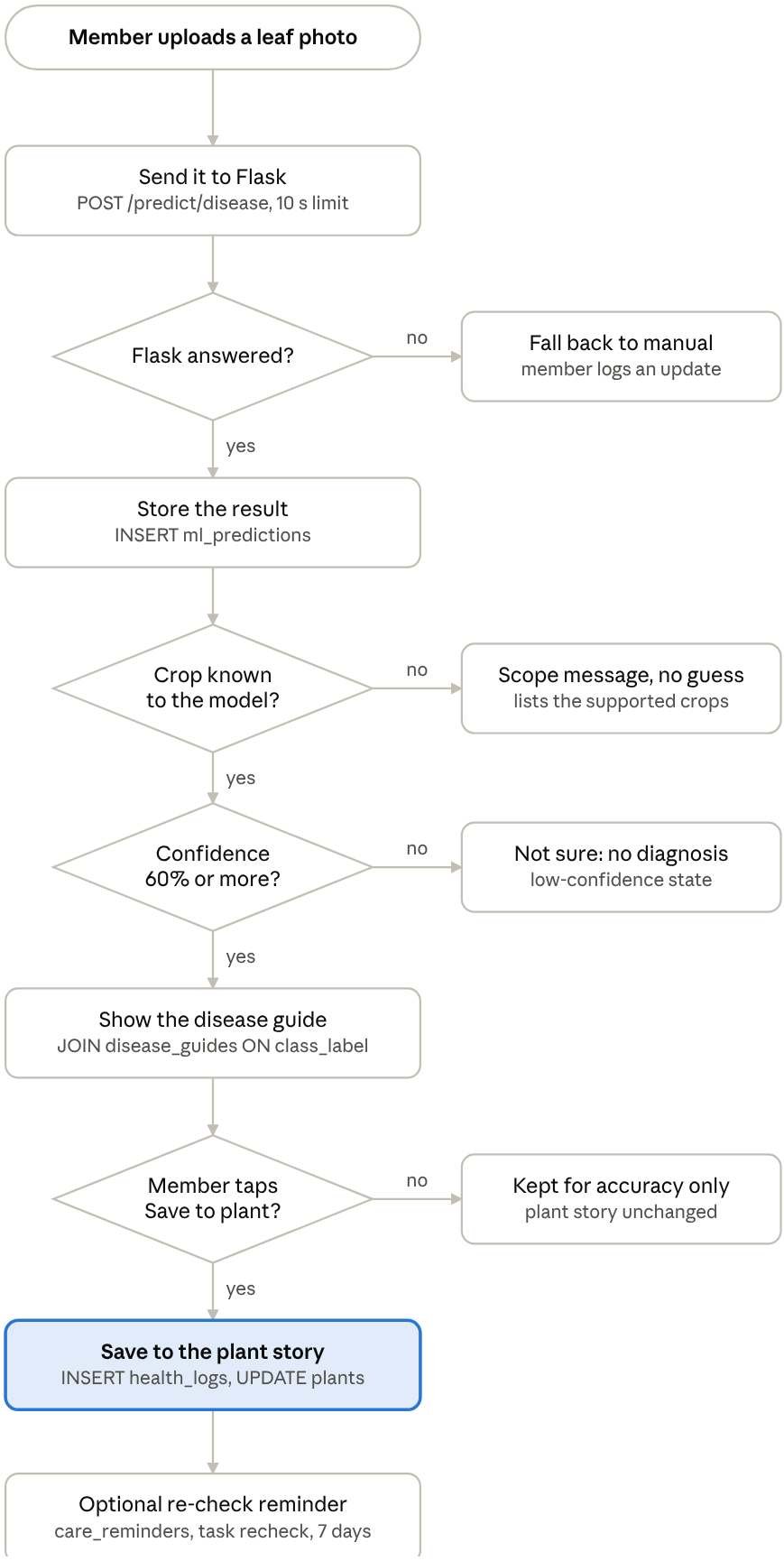


Flowchart 3 · care reminder cycle

No background scheduler is needed: the Dashboard simply asks for rows whose `next_due` is today or earlier each time it loads.

4. Health Check

A diagnosis reaches the plant story only when the member saves it

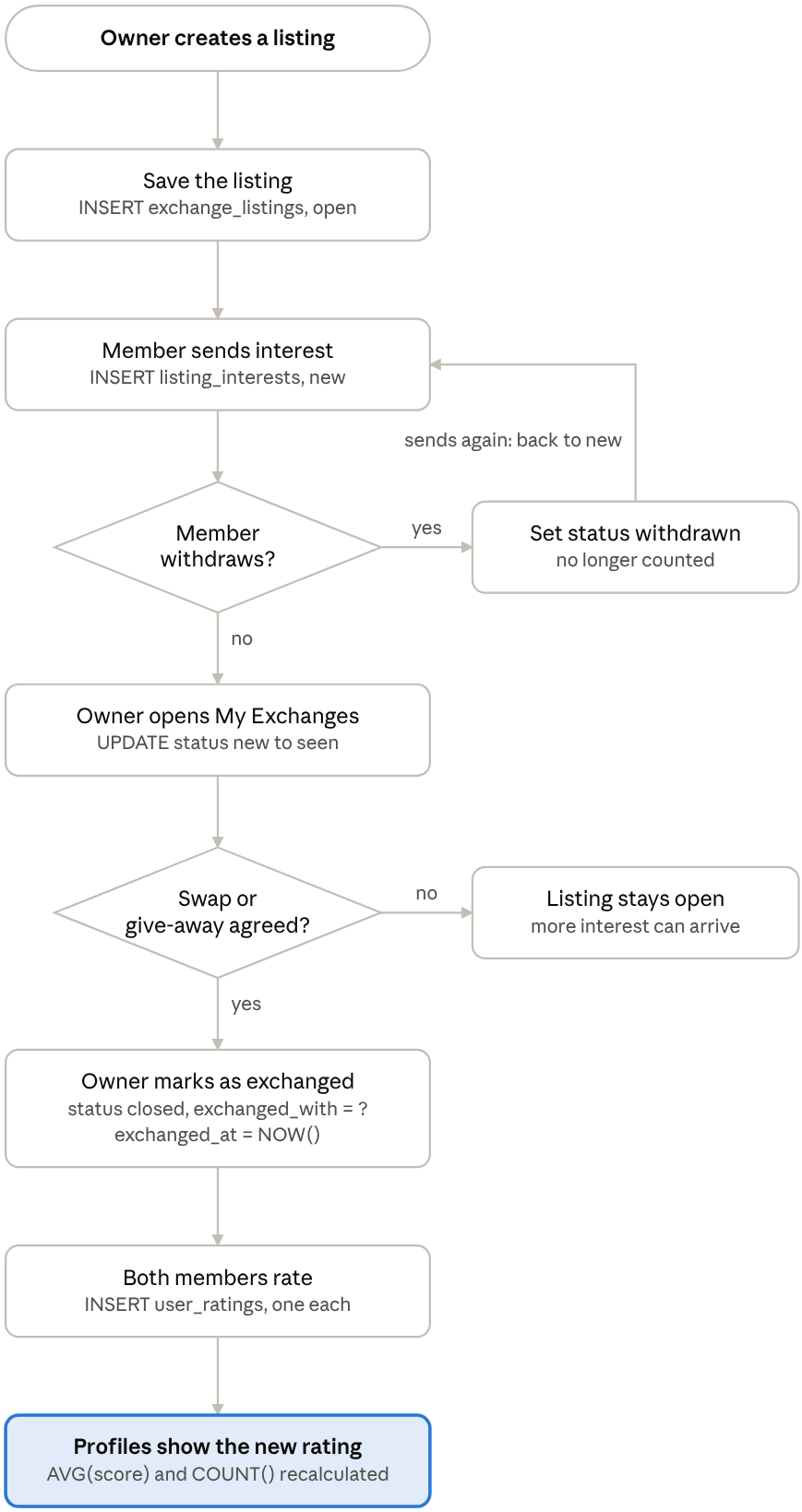


Flowchart 4 · Health Check

Every result is stored in `ml_predictions`, even when it is not saved, so accuracy can be measured for the report. Saving writes the log, updates `plants.health_status` and links the prediction in one transaction.

5. Exchange

A listing closes once, then each member can rate the other once



Flowchart 5 · exchange lifecycle

The UNIQUE key on (listing_id, user_id) keeps one interest row per member, so sending again flips the status back to new instead of adding a row. The UNIQUE key on (rater_id, listing_id) stops a second rating of the same exchange.

SQL schema script

Save this as `database/schema.sql`. It creates all 14 tables, 22 foreign keys and every index, and it was tested on MariaDB 10.11 (the engine XAMPP ships): it runs twice in a row without errors, refuses a second rating of the same exchange, refuses a score of 9, and deleting a user removes their plants and ratings.

```
-- PlantFolio schema (CORE-01): 14 tables, InnoDB, utf8mb4. Re-runs cleanly
on an empty database.
SET NAMES utf8mb4;
SET FOREIGN_KEY_CHECKS = 0;
DROP TABLE IF EXISTS user_ratings, listing_interests, listing_photos,
exchange_listings,
  disease_guides, health_logs, ml_predictions, care_reminders, plant_likes,
plant_photos,
  plants, plant_species, collections, users;
SET FOREIGN_KEY_CHECKS = 1;

CREATE TABLE users (
  id                  INT UNSIGNED     NOT NULL AUTO_INCREMENT,
  username            VARCHAR(30)      NOT NULL,
  full_name           VARCHAR(100)     NOT NULL,
  email               VARCHAR(255)     NOT NULL,
  phone               VARCHAR(20)      NULL,
  password_hash       VARCHAR(255)     NOT NULL,
  bio                 VARCHAR(500)     NULL,
  location            VARCHAR(100)     NULL,
  avatar              VARCHAR(255)     NULL,
  new_plants_public   TINYINT(1)       NOT NULL DEFAULT 1,
  show_listing_area   TINYINT(1)       NOT NULL DEFAULT 1,
  dashboard_reminders TINYINT(1)       NOT NULL DEFAULT 1,
  email_reminders     TINYINT(1)       NOT NULL DEFAULT 0,
  failed_logins       TINYINT UNSIGNED NOT NULL DEFAULT 0,
  locked_until        DATETIME         NULL,
  created_at          DATETIME         NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (id),
```

```
  UNIQUE KEY uq_users_username (username),
  UNIQUE KEY uq_users_email (email)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE collections (
  id         INT UNSIGNED NOT NULL AUTO_INCREMENT,
  user_id    INT UNSIGNED NOT NULL,
  name       VARCHAR(60)  NOT NULL,
  is_public  TINYINT(1)   NOT NULL DEFAULT 1,
  created_at DATETIME     NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (id),
  UNIQUE KEY uq_collections_user_name (user_id, name),
  CONSTRAINT fk_collections_user FOREIGN KEY (user_id) REFERENCES users (id)
ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE plant_species (
  id                     INT UNSIGNED      NOT NULL AUTO_INCREMENT,
  common_name            VARCHAR(100)      NOT NULL,
  scientific_name        VARCHAR(150)      NOT NULL,
  photo                  VARCHAR(255)      NULL,
  light                  VARCHAR(100)      NULL,
  water                  VARCHAR(150)      NULL,
  humidity               VARCHAR(60)       NULL,
  pet_safe               TINYINT(1)        NULL,
  water_interval_days    SMALLINT UNSIGNED NOT NULL DEFAULT 7,
  fertilise_interval_days SMALLINT UNSIGNED NOT NULL DEFAULT 30,
  notes                  TEXT              NULL,
  PRIMARY KEY (id),
  UNIQUE KEY uq_species_scientific (scientific_name),
  KEY idx_species_common (common_name)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE plants (
  id            INT UNSIGNED NOT NULL AUTO_INCREMENT,
  user_id       INT UNSIGNED NOT NULL,
  collection_id INT UNSIGNED NULL,
  species_id    INT UNSIGNED NULL,
  nickname      VARCHAR(60)  NOT NULL,
  photo         VARCHAR(255) NULL,
  acquired_on   DATE         NULL,
  health_status ENUM('thriving','stable','needs_attention','sick') NOT NULL
DEFAULT 'stable',
```

```
  is_public     TINYINT(1)   NOT NULL DEFAULT 1,
  created_at    DATETIME     NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (id),
  KEY idx_plants_user (user_id),
  KEY idx_plants_collection (collection_id),
  KEY idx_plants_species (species_id),
  KEY idx_plants_health (health_status),
  KEY idx_plants_public (is_public),
  CONSTRAINT fk_plants_user       FOREIGN KEY (user_id)       REFERENCES
users (id)         ON DELETE CASCADE,
  CONSTRAINT fk_plants_collection FOREIGN KEY (collection_id) REFERENCES
collections (id)   ON DELETE SET NULL,
  CONSTRAINT fk_plants_species    FOREIGN KEY (species_id)    REFERENCES
plant_species (id) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE plant_photos (
  id         INT UNSIGNED NOT NULL AUTO_INCREMENT,
  plant_id   INT UNSIGNED NOT NULL,
  path       VARCHAR(255) NOT NULL,
  created_at DATETIME     NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (id),
  KEY idx_plant_photos_plant (plant_id),
  CONSTRAINT fk_plant_photos_plant FOREIGN KEY (plant_id) REFERENCES plants
(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE plant_likes (
  id         INT UNSIGNED NOT NULL AUTO_INCREMENT,
  plant_id   INT UNSIGNED NOT NULL,
  user_id    INT UNSIGNED NOT NULL,
  created_at DATETIME     NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (id),
  UNIQUE KEY uq_plant_likes (plant_id, user_id),
  KEY idx_plant_likes_user (user_id),
  CONSTRAINT fk_plant_likes_plant FOREIGN KEY (plant_id) REFERENCES plants
(id) ON DELETE CASCADE,
  CONSTRAINT fk_plant_likes_user  FOREIGN KEY (user_id)  REFERENCES users
(id)  ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE care_reminders (
  id            INT UNSIGNED      NOT NULL AUTO_INCREMENT,
```

```
  plant_id      INT UNSIGNED      NOT NULL,
  task          ENUM('water','fertilise','recheck') NOT NULL,
  interval_days SMALLINT UNSIGNED NOT NULL,
  last_done     DATE              NULL,
  next_due      DATE              NOT NULL,
  PRIMARY KEY (id),
  UNIQUE KEY uq_reminders_plant_task (plant_id, task),
  KEY idx_reminders_next_due (next_due),
  CONSTRAINT chk_reminders_interval CHECK (interval_days > 0),
  CONSTRAINT fk_reminders_plant FOREIGN KEY (plant_id) REFERENCES plants (id)
ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE ml_predictions (
  id              INT UNSIGNED  NOT NULL AUTO_INCREMENT,
  user_id         INT UNSIGNED  NOT NULL,
  plant_id        INT UNSIGNED  NULL,
  type            ENUM('species','disease') NOT NULL,
  predicted_label VARCHAR(150)  NOT NULL,
  confidence      DECIMAL(5,4)  NOT NULL,
  alternatives    JSON          NULL,
  image_path      VARCHAR(255)  NOT NULL,
  upload_name     VARCHAR(255)  NULL,
  upload_size     INT UNSIGNED  NULL,
  model_version   VARCHAR(50)   NOT NULL,
  created_at      DATETIME      NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (id),
  KEY idx_predictions_user (user_id),
  KEY idx_predictions_plant (plant_id),
  KEY idx_predictions_label (predicted_label),
  CONSTRAINT chk_predictions_confidence CHECK (confidence BETWEEN 0 AND 1),
  CONSTRAINT fk_predictions_user  FOREIGN KEY (user_id)  REFERENCES users
(id)  ON DELETE CASCADE,
  CONSTRAINT fk_predictions_plant FOREIGN KEY (plant_id) REFERENCES plants
(id) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE health_logs (
  id            INT UNSIGNED NOT NULL AUTO_INCREMENT,
  plant_id      INT UNSIGNED NOT NULL,
  prediction_id INT UNSIGNED NULL,
  status        ENUM('thriving','stable','needs_attention','sick') NOT NULL,
  source        ENUM('manual','ml') NOT NULL DEFAULT 'manual',
```

```
  note          VARCHAR(500) NULL,
  photo         VARCHAR(255) NULL,
  logged_at     DATETIME     NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (id),
  KEY idx_health_logs_plant_date (plant_id, logged_at),
  KEY idx_health_logs_prediction (prediction_id),
  CONSTRAINT fk_health_logs_plant      FOREIGN KEY (plant_id)      REFERENCES
plants (id)         ON DELETE CASCADE,
  CONSTRAINT fk_health_logs_prediction FOREIGN KEY (prediction_id) REFERENCES
ml_predictions (id) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE disease_guides (
  id          INT UNSIGNED NOT NULL AUTO_INCREMENT,
  class_label VARCHAR(100) NOT NULL,
  crop        VARCHAR(50)  NOT NULL,
  cause       VARCHAR(255) NULL,
  symptoms    TEXT         NULL,
  steps       JSON         NULL,
  status      ENUM('thriving','stable','needs_attention','sick') NOT NULL,
  PRIMARY KEY (id),
  UNIQUE KEY uq_disease_guides_label (class_label)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE exchange_listings (
  id              INT UNSIGNED NOT NULL AUTO_INCREMENT,
  user_id         INT UNSIGNED NOT NULL,
  plant_id        INT UNSIGNED NULL,
  species_id      INT UNSIGNED NULL,
  title           VARCHAR(100) NOT NULL,
  description     TEXT         NULL,
  type            ENUM('swap','free','either') NOT NULL DEFAULT 'swap',
  looking_for     VARCHAR(150) NULL,
  location        VARCHAR(100) NOT NULL,
  size            VARCHAR(60)  NULL,
  plant_condition VARCHAR(100) NULL,
  photo           VARCHAR(255) NOT NULL,
  share_email     TINYINT(1)   NOT NULL DEFAULT 1,
  share_phone     TINYINT(1)   NOT NULL DEFAULT 0,
  approx_area     TINYINT(1)   NOT NULL DEFAULT 1,
  status          ENUM('open','closed') NOT NULL DEFAULT 'open',
  exchanged_with  INT UNSIGNED NULL,
  exchanged_at    DATETIME     NULL,
```

```
  created_at      DATETIME     NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (id),
  KEY idx_listings_user (user_id),
  KEY idx_listings_plant (plant_id),
  KEY idx_listings_species (species_id),
  KEY idx_listings_exchanged_with (exchanged_with),
  KEY idx_listings_status_created (status, created_at),
  CONSTRAINT fk_listings_user      FOREIGN KEY (user_id)        REFERENCES
users (id)         ON DELETE CASCADE,
  CONSTRAINT fk_listings_plant     FOREIGN KEY (plant_id)       REFERENCES
plants (id)        ON DELETE SET NULL,
  CONSTRAINT fk_listings_species   FOREIGN KEY (species_id)     REFERENCES
plant_species (id) ON DELETE SET NULL,
  CONSTRAINT fk_listings_exchanged FOREIGN KEY (exchanged_with) REFERENCES
users (id)         ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE listing_photos (
  id         INT UNSIGNED NOT NULL AUTO_INCREMENT,
  listing_id INT UNSIGNED NOT NULL,
  path       VARCHAR(255) NOT NULL,
  created_at DATETIME     NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (id),
  KEY idx_listing_photos_listing (listing_id),
  CONSTRAINT fk_listing_photos_listing FOREIGN KEY (listing_id) REFERENCES
exchange_listings (id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

CREATE TABLE listing_interests (
  id         INT UNSIGNED NOT NULL AUTO_INCREMENT,
  listing_id INT UNSIGNED NOT NULL,
  user_id    INT UNSIGNED NOT NULL,
  message    VARCHAR(500) NULL,
  status     ENUM('new','seen','withdrawn') NOT NULL DEFAULT 'new',
  created_at DATETIME     NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (id),
  UNIQUE KEY uq_interests_listing_user (listing_id, user_id),
  KEY idx_interests_user (user_id),
  CONSTRAINT fk_interests_listing FOREIGN KEY (listing_id) REFERENCES
exchange_listings (id) ON DELETE CASCADE,
  CONSTRAINT fk_interests_user    FOREIGN KEY (user_id)    REFERENCES users
(id)             ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

```
CREATE TABLE user_ratings (
  id         INT UNSIGNED     NOT NULL AUTO_INCREMENT,
  rater_id   INT UNSIGNED     NOT NULL,
  rated_id   INT UNSIGNED     NOT NULL,
  listing_id INT UNSIGNED     NOT NULL,
  score      TINYINT UNSIGNED NOT NULL,
  comment    VARCHAR(500)     NULL,
  created_at DATETIME         NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (id),
  UNIQUE KEY uq_ratings_rater_listing (rater_id, listing_id),
  KEY idx_ratings_rated (rated_id),
  KEY idx_ratings_listing (listing_id),
  CONSTRAINT chk_ratings_score CHECK (score BETWEEN 1 AND 5),
  CONSTRAINT fk_ratings_rater   FOREIGN KEY (rater_id)   REFERENCES users
(id)             ON DELETE CASCADE,
  CONSTRAINT fk_ratings_rated   FOREIGN KEY (rated_id)   REFERENCES users
(id)             ON DELETE CASCADE,
  CONSTRAINT fk_ratings_listing FOREIGN KEY (listing_id) REFERENCES
exchange_listings (id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

One rule is enforced in PHP rather than SQL: a member cannot rate themselves (`rated_id <> rater_id`). MySQL 8 refuses a CHECK on a column whose foreign key has ON DELETE CASCADE, so leaving it out keeps the script portable.

Indexes, normalisation and key queries

The schema is in third normal form. The one deliberate copy is `plants.health_status` , which repeats the newest health log's status so My Garden and the Dashboard can filter without reading every log; PHP updates it in the same transaction that inserts the log.

Normalisation check

- **1NF:** every column holds one value. Extra photos live in `plant_photos` and `listing_photos` , not in comma lists. The two JSON columns (`alternatives` , `steps`) are always read whole and never searched, an accepted exception.
- **2NF:** every table has a single-column primary key, so no column can depend on part of a key.
- **3NF:** nothing depends on a non-key column. A listing never stores its owner's name or rating; those are joined from `users` and `user_ratings` .

Index	Speeds up
care_reminders (next_due)	Dashboard "due today" and "overdue" list
plants (health_status)	Dashboard attention list, My Garden filter
plants (is_public)	Explore search, Public Profile
health_logs (plant_id, logged_at)	Plant story, newest first
exchange_listings (status, created_at)	Marketplace open listings, Newest sort
ml_predictions (predicted_label)	Joining a result to its species or disease guide
UNIQUE keys on users, plant_likes, listing_interests, user_ratings, care_reminders	Stop duplicate accounts, likes, interests, ratings and reminders

Key queries (all run through PDO prepared statements; `?` is the bound value)

```
-- Dashboard: care due today or overdue for the logged-in member
SELECT r.id, r.task, r.next_due, p.nickname, p.photo,
       DATEDIFF(CURDATE(), r.next_due) AS days_overdue
FROM care_reminders r
JOIN plants p ON p.id = r.plant_id
WHERE p.user_id = ? AND r.next_due <= CURDATE()
ORDER BY r.next_due;

-- Mark done: the ownership check is in the JOIN, so another member's id
changes nothing
UPDATE care_reminders r
JOIN plants p ON p.id = r.plant_id
SET r.last_done = CURDATE(),
    r.next_due  = CURDATE() + INTERVAL r.interval_days DAY
WHERE r.id = ? AND p.user_id = ?;

-- Save a Health Check result to the plant story (one transaction)
START TRANSACTION;
INSERT INTO health_logs (plant_id, prediction_id, status, source, note,
photo)
VALUES (?, ?, ?, 'ml', ?, ?);
UPDATE plants SET health_status = ? WHERE id = ? AND user_id = ?;
UPDATE ml_predictions SET plant_id = ? WHERE id = ? AND user_id = ?;
```

```
COMMIT;

-- Marketplace cards: open listings with owner rating and interest count
SELECT l.*, u.username, u.full_name, u.avatar,
       COALESCE(p.species_id, l.species_id) AS species_id,
       (SELECT ROUND(AVG(score), 1) FROM user_ratings WHERE rated_id =
l.user_id) AS rating_avg,
       (SELECT COUNT(*) FROM user_ratings WHERE rated_id = l.user_id) AS
rating_count,
       (SELECT COUNT(*) FROM listing_interests i
         WHERE i.listing_id = l.id AND i.status <> 'withdrawn') AS
interest_count
FROM exchange_listings l
JOIN users u ON u.id = l.user_id
LEFT JOIN plants p ON p.id = l.plant_id
WHERE l.status = 'open'
ORDER BY l.created_at DESC;
```

Open questions for DES-07

- Listing cards show a distance ("3 km"). That needs coordinates for each town: a small `towns` table (name, latitude, longitude), or drop the distance.
- The login form has a "Remember me" box. A longer session cookie needs no table; a secure remember-me token needs a `remember_tokens` table.
- Deleting a member also deletes the reviews they wrote, which changes other members' averages. The alternative is to keep the review and show "Former member".
- `disease_guides` can be this table or a JSON file next to the model's class list (DES-07 / DES-10).